

Cycle Universality for a Single Thorp Shuffle

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Abstract

We study the cycles of one physical Thorp shuffle on $N = 2^d$ states, equivalently a uniformly random binary noncoalescing feedback shift register. For every fixed number of prescribed cycles, we prove uniform joint factorial-moment asymptotics when their lengths are at least $4d$ and their total length is at most N^β , for fixed $\beta < 2/3$. No restriction is imposed on the ratios of the cycle lengths. These estimates yield a joint functional Gaussian limit for cycle counts and logarithmic truncated permutation order on every fixed exponent interval $[0, \alpha]$, $\alpha < 2/3$. They also give Poisson approximation on sets of bounded harmonic mass, a scale-invariant Poisson cycle process, and Dickman and uniform limits for truncated cycle mass and the largest cycle below a cutoff. A separate argument gives a quantitative Poisson approximation for the entire initial cycle-count vector below the birthday scale. The proof combines weighted cyclic words, empirical-type entropy, marked switching, longest-last forced closure, and a middle-first bridge. The feedback function is sampled once and reused. The full permutation-order central limit theorem is not proved.

1 Introduction and main results

A random permutation can have universal cycle statistics long before its full law resembles the uniform measure. We study this distinction for one physical Thorp shuffle on $N = 2^d$ positions. In binary coordinates its permutation is

$$T_f(x_1, \dots, x_d) = (x_2, \dots, x_d, x_1 \oplus f(x_2, \dots, x_d)), \quad f : \mathbb{F}_2^{d-1} \longrightarrow \mathbb{F}_2,$$

where all $N/2$ values of f are independent fair bits. This is the binary noncoalescing feedback-shift-register model. The feedback function is sampled *once*; following a cycle reuses its values, rather than supplying an independent table at every iterate.

Thorp introduced the shuffle in his study of nonrandom shuffling and Faro [8]. For this feedback model, Arratia, Canfield, and Hales [2] proved the Poisson–Dirichlet limit for the ranked macroscopic cycle lengths. Stark [7] proved exact independent-binomial laws for sufficiently short cycles. Neither a macroscopic weak limit nor short-cycle independence determines the joint statistics of cycles at intermediate lengths. The present paper establishes uniform factorial moments across that intermediate regime, with no comparability assumption on the prescribed lengths.

Write $C_r = C_r(T_f)$ for the number of r -cycles, $u(x)$ for the last $d-1$ bits of a state, and $\bar{x}$ for the state obtained by flipping its first bit. We use $\mathcal{L}(Y)$ for a law and $d_{\text{TV}}(\mu, \nu) = \sup_A |\mu(A) - \nu(A)|$. Entropies and $\log_2$ are in bits; other logarithms are natural. The notation $(z)_a$ denotes a falling factorial, and the least common multiple of an empty set is one.

Our main estimates lead to a truncated functional Erdős–Turán law, complementing the classical permutation-order theorem [5] and one-shuffle universality results for riffle and shelf models [6]. They also give Poisson approximation on sets of bounded harmonic mass, a scale-invariant Poisson

cycle process, and a Dickman law for the mass in truncated cycles. A separate quantitative estimate treats the full initial cycle-count vector below the birthday scale. The proof uses weighted cyclic words, empirical-type entropy, marked feedback switches, and forced closures. It uses neither an independent-round mixing theorem nor the macroscopic Poisson–Dirichlet limit.

1.1 Joint cycle moments

For a list of positive lengths $\mathbf{r} = (r_1, \dots, r_k)$, put

$$L = \sum_{i=1}^k r_i, \quad a_m = \#\{i : r_i = m\}, \quad Q_d(\mathbf{r}) = \left(\prod_{i=1}^k r_i \right) \mathbb{E} \prod_{m \geq 1} (C_m)_{a_m}, \quad Q_d(\emptyset) = 1.$$

Fix an integer $k \geq 1$ and a real $0 < \beta < 2/3$. Set

$$p_\beta = \frac{2}{2 - 3\beta}, \quad P_{k,\beta}(d) = d^{4k+1} + d(d \log_2 d)^{kp_\beta}.$$

Theorem 1.1 (Uniform cycle factorial moments). *Fix $k \geq 1$ and $0 < \beta < 2/3$. For all sufficiently large d , depending on k and β , every list of positive lengths with $L \leq N^\beta$ satisfies*

$$Q_d(\mathbf{r}) \leq 1 + C_{k,\beta} P_{k,\beta}(d) \frac{L}{N}. \quad (1.1)$$

If additionally $r_i \geq 4d$ for every i , then

$$1 - C_k \frac{dL}{N} \leq Q_d(\mathbf{r}) \leq 1 + C_{k,\beta} P_{k,\beta}(d) \frac{L}{N}. \quad (1.2)$$

Constants are independent of all ratios r_i/r_j . In particular,

$$rs \mathbb{E}[C_r(C_s - \mathbf{1}_{\{r=s\}})] = 1 + O_\beta \left(P_{2,\beta}(d) \frac{r+s}{N} \right)$$

uniformly for $r, s \geq 4d$ and $r+s \leq N^\beta$. Equal lengths are covered by the falling factorial, so the cycles are required to be distinct.

The displayed growth factor is deliberately not optimized. The conclusion is uniform with exponentially small error in d on every fixed exponent range.

1.2 Functional consequence

Theorem 1.2 (Truncated functional Erdős–Turán law). *Fix $0 < \alpha < 2/3$, put $\ell = \log N$, and define, for $0 \leq u \leq \alpha$,*

$$K_d(u) = \sum_{r \leq \lfloor N^u \rfloor} C_r, \quad O_d(u) = \log \text{lcm}\{r \leq \lfloor N^u \rfloor : C_r > 0\}.$$

Then, in $D([0, \alpha], \mathbb{R}^2)$ with its Skorokhod J_1 topology,

$$\left(\frac{K_d(u) - u\ell}{\sqrt{\ell}}, \frac{O_d(u) - \frac{1}{2}u^2\ell^2}{\sqrt{\ell^3/3}} \right)_{0 \leq u \leq \alpha} \Longrightarrow \left(W(u), \sqrt{3} \int_0^u t dW(t) \right)_{0 \leq u \leq \alpha}. \quad (1.3)$$

Here W is one standard Brownian motion, shared by both coordinates. In particular, for $b = \lfloor N^\alpha \rfloor$,

$$\frac{\log \text{lcm}\{r \leq b : C_r > 0\} - \frac{1}{2}(\log b)^2}{\sqrt{(\log b)^3/3}} \Longrightarrow \mathcal{N}(0, 1). \quad (1.4)$$

The proof of the functional consequence, including tightness, is given in Sections 8–9. The parameters k, β, α are fixed before $d \rightarrow \infty$; no uniform assertion as $k \rightarrow \infty$ or $\beta \uparrow 2/3$ is made.

2 Exact cycle measures and weighted words

Choose k independent uniform roots, with replacement, independently of f . Let $E_{\mathbf{r}}$ be the event that their cycles are distinct and have the prescribed lengths. Then the finite measure

$$\mu_{\mathbf{r}} = N^k \mathbb{P}_0(\cdot, E_{\mathbf{r}})$$

has total mass $Q_d(\mathbf{r})$. An unrooted ordered cycle family has the same weight at every choice of roots on its cycles.

Equivalently, prescribe k distinguished cyclic binary words with these lengths. Let A_d require all their length- d windows to be distinct. A context is a length- $(d-1)$ window immediately following the initial bit of a state. On A_d , a context has at most two occurrences. If a context occurs twice, the two preceding bits differ and the two following bits differ, so the feedback requirements are consistent. If $m = L - v$, with v distinct contexts, the family occurs with probability 2^{-v} . Therefore

$$Q_d^{(m)}(\mathbf{r}) = 2^{m-L} \# \{\text{valid rooted word families with } m \text{ repeated contexts}\}. \quad (2.1)$$

This identity counts all repeats, internal or between cycles. No context-simplicity condition is imposed.

3 A global entropy tail for a fixed number of words

3.1 Type counting, including short component words

For a cyclic binary word of length n , let H_h be the entropy of its empirical h -block distribution and $\delta_h = H_h - H_{h-1}$. A fixed h -type contains at most

$$n 2^{n\delta_h} \quad (3.1)$$

rooted words. For $h \leq n$, use the empirical order- $(h-1)$ Markov chain: every word of the type has probability at least $n^{-1}2^{-n\delta_h}$, since its initial block has mass at least $1/n$ and omitting wraparound transition factors only increases the product.

The bound is also valid for $h > n$, a point needed for widely separated lengths. An h -block then contains the entire cyclic word. A fixed h -type determines its necklace, so at most n rooted words have the type. Also $\delta_h = 0$ once $h-1 \geq n$.

Writing $M = 2^h$, the number of empirical types is at most

$$\binom{n+M-1}{M-1} \leq [e(1+n/M)]^M. \quad (3.2)$$

Indeed, the nonnegative block counts sum to n . Stars and bars gives the first bound in (3.2); then $\binom{n+M-1}{M-1} \leq \binom{n+M}{M} \leq [e(n+M)/M]^M$. No assumption that different component types are independent is used.

3.2 Mixture entropy

For a valid family, sample a position uniformly among its L positions and read the resulting stationary binary process. All d -blocks are distinct, so

$$H_d = \log_2 L, \quad H_d - H_{d-1} = \frac{2m}{L}.$$

Stationary conditional entropies decrease with block length. Thus

$$\delta_h \leq \frac{H_h}{h} \leq \frac{\log_2 L - 2m(d-h)/L}{h}. \quad (3.3)$$

Let J identify the component circle and let X, Y be the preceding $h-1$ bits and the final bit. Then

$$L\delta_h^{\text{mix}} - \sum_i r_i \delta_h(w_i) = LI(Y; J | X) \geq 0. \quad (3.4)$$

This remains valid for any number of component circles and arbitrary lengths. Applying (3.1)–(3.4), and bounding the rooted prefactor by L^k , yields

$$\log_2 Q_d^{(m)}(\mathbf{r}) \leq \frac{Lt}{h} + kM \log_2[e(1 + L/M)] + k \log_2 L - \kappa m, \quad (3.5)$$

where

$$t = \log_2 L - h, \quad \kappa = \frac{2d - 3h}{h}.$$

3.3 The sparse threshold

Suppose $d^4 \leq L \leq N^\beta$, and put $q = \log_2 d$ and

$$h = \lfloor \log_2 L - \log_2 d - \log_2 q \rfloor.$$

For sufficiently large d , $h \geq 2$ and $h \leq \beta d$,

$$\kappa \geq (2 - 3\beta)/\beta > 0, \quad kM \log_2[e(1 + L/M)] \leq 2kL/d.$$

Let

$$F = Lt/h + 2kL/d + k \log_2 L, \quad m_0 = \left\lceil \frac{F + L/d}{\kappa} \right\rceil.$$

Summing (3.5) gives

$$D_{\mathbf{r}} := \sum_{m \geq m_0} Q_d^{(m)}(\mathbf{r}) \leq C_\beta 2^{-L/d}. \quad (3.6)$$

Furthermore,

$$\frac{2dm_0}{L} \leq p_\beta(\log_2 d + \log_2 \log_2 d) + O_{k,\beta}(1). \quad (3.7)$$

For clarity, the algebra behind (3.7) is

$$\frac{2dm_0}{L} \leq \frac{2d}{2d - 3h} \left[t + (2k + 1)\frac{h}{d} + \frac{kh \log_2 L}{L} \right] + \frac{2d}{L}.$$

The leading multiplier is at most p_β ; $t \in [\log_2 d + \log_2 \log_2 d, \log_2 d + \log_2 \log_2 d + 1)$, $h/d \leq \beta$, and $h \log_2 L/L = O_\beta(d^{-2})$ because $L \geq d^4$. Also, summation in (3.6) uses the finite positive lower bound on κ : $\sum_{m \geq m_0} 2^{F - \kappa m} \leq 2^{-L/d}/(1 - 2^{-\kappa})$.

When $r = \max_i r_i$, $L/r \leq k$. Hence the closing penalty at the longest cycle's scale satisfies

$$2^{2dm_0/r} \leq C_{k,\beta}(d \log_2 d)^{kp_\beta}. \quad (3.8)$$

For $L < d^4$ no dense decomposition is necessary. Stationarity also gives

$$m \leq \frac{L \log_2 L}{2d},$$

so, for any $m_{\text{new}} \leq m$,

$$2^{2dm_{\text{new}}/r} \leq L^{L/r} \leq L^k \leq d^{4k}. \quad (3.9)$$

In either case the needed sparse penalty is bounded by

$$H_{k,\beta}(d) = C_{k,\beta} \left[d^{4k} + (d \log_2 d)^{kp_\beta} \right]. \quad (3.10)$$

It depends on k , not on the ratios between the component lengths.

4 Marked stopped collisions with earlier cycles present

Let $\mathbf{a} = (r_1, \dots, r_{k-1})$, $P = \sum_{i=1}^{k-1} r_i$, and let $\mu_{\mathbf{a}}$ be its scaled ordinary cycle measure. Conditional on the recorded rooted cycles, feedback values at all unqueried contexts remain independent fair bits: specifying the recorded cycles imposes exactly the feedback constraints along them.

Adjoin a new independent uniform root X and run its ordinary trajectory for n steps. Retain it only if its first $n+1$ states are distinct and outside all the earlier cycles. Let J_{int} count repeated contexts inside that prefix and J_{cross} count prefix queries using contexts of earlier cycles. The latter statistic may be bounded after dropping the retention restriction.

4.1 Internal contacts and marked switching

For $h > 0$, the conjugate endpoint count must include the condition $L_f(y) > h$. On an internal conjugate encounter of the new surviving path, both states of the conjugate pair lie in a cycle disjoint from all earlier cycles. Their context is therefore unused by every earlier cycle. Flipping its feedback bit preserves all earlier marked cycles and splits off an h -cycle.

The resulting marked configurations are a subset of those counted by $Q_d(\mathbf{a}, h)$. Stationarity of the new uniform root then gives

$$\mathbb{E}_{\mu_{\mathbf{a}}, X} J_{\text{int}} \leq \frac{1}{N} \sum_{h=1}^{n-1} (n-h) Q_d(\mathbf{a}, h). \quad (4.1)$$

This is not a conditional-uniformity assertion about f under $\mu_{\mathbf{a}}$. It follows from the feedback-flip bijection while keeping the previously selected cycles marked. The survival and disjointness restrictions ensure the flip leaves those marks unchanged. Sorting the list $(\mathbf{a}, h)$ is permitted because Q is symmetric.

Here is the normalization and bijection in (4.1). Write $\mathcal{C}_f(y)$ for the cycle containing y and A for the union of the previously marked cycles. Feedback-bit switching gives

$$\int \mu_{\mathbf{a}}(d(f, \text{marks})) \sum_{\substack{y: T_f^h y = \bar{y}, L_f(y) > h \\ \mathcal{C}_f(y) \cap A = \emptyset}} 1 \leq Q_d(\mathbf{a}, h). \quad (4.1a)$$

To see this, flip the bit at $u(y)$. In the new permutation the cycle containing $\bar{y}$ has length h , and y is outside it. All earlier marked cycles, including their roots, are unchanged. The map is injective: from the new marked h -cycle rooted at $\bar{y}$, recover y , flip the same bit, and recover the original configuration. Counting all marked h -cycles can only enlarge the image. The uniform feedback law is invariant under this flip. The factor h in $Q_d(\mathbf{a}, h)$ counts its possible new roots, including when h equals one of the earlier lengths.

For a retained prefix and query times $i < j$, take $y = T_f^i X$ and $h = j - i$. Conditional on f and the earlier marks, y is uniform. The retained prefix is in a cycle disjoint from A ; disjointness of a prefix from A implies this because A is a union of complete cycles. It also has $L_f(y) > n > h$. Dropping the stronger retention conditions and applying (4.1a) gives $Q_d(\mathbf{a}, h)/N$ for each pair. There are $n - h$ query pairs with separation h , proving (4.1). No unconditioned stationarity assertion is substituted for this marked calculation.

4.2 Cross contacts

For fixed f and fixed earlier cycles, every iterate of X is uniform because T_f is a permutation. The earlier cycles use at most P contexts. Each context has exactly two possible preceding states, so

$$\mathbb{E}_{\mu_{\mathbf{a}}, X} J_{\text{cross}} \leq \frac{2nP}{N} Q_d(\mathbf{a}). \quad (4.2)$$

On a retained prefix, each context occurs at most twice across its query states and the earlier cycles: a context has only two preceding states, and all these states are distinct. Hence a prefix query cannot simultaneously repeat an earlier prefix query and a context in the marked cycles. Thus the reused-query count is exactly $J = J_{\text{int}} + J_{\text{cross}}$ on retained histories. Both quantities are set to zero on discarded histories; only the cross bound is enlarged by dropping retention.

Consequently the ordinary open-repeat statistic J satisfies

$$\mathbb{E}J \leq \frac{1}{N} \sum_{h=1}^{n-1} (n-h) Q_d(\mathbf{a}, h) + \frac{2nP}{N} Q_d(\mathbf{a}). \quad (4.3)$$

The first term involves the same number of cycles but strictly smaller total length. The second involves one fewer cycle. This is the induction structure.

5 Longest-last forced closure

Sort $r_1 \leq \dots \leq r_k = r$. Keep the earlier cycles under $\mu_{\mathbf{a}}$. Starting from a new uniform root, expose $r - d$ ordinary open steps, then force the final d outputs to equal the initial state. Never change an already prescribed feedback value. Success requires first return at exactly r and disjointness from all earlier cycles. Complete unqueried feedback values fairly.

Operationally, first sample the *recorded* marked-cycle histories from the projection of $\mu_{\mathbf{a}}$, which is a finite measure. Sample only the still-unqueried feedback values as the new trajectory needs them, and complete the remaining values at the end. This avoids treating the biased law after an earlier forced closure as an ordinary feedback law. Every previously selected cycle constrains only its own queried contexts.

Denote this finite successful measure by $\nu_{\mathbf{a}; r}$. Its mass is at most $Q_d(\mathbf{a})$, not necessarily at most one. Let K count closing queries whose feedback value has already been fixed strictly before that query is made: by an earlier marked cycle, by the current open path, or by an earlier query in the

current closing segment. In particular, two occurrences of a context within the closing segment contribute one to K . On successful histories,

$$d\nu_{\mathbf{a};r} = 2^{-K} d\mu_{\mathbf{a},r}, \quad (5.1)$$

For an atom with v_{old} old contexts and v contexts in the completed family, the scaled ordinary word weight is 2^{-v} . The forced construction samples the new root with factor N^{-1} and samples only $v - v_{\text{old}} - (d - K)$ additional feedback bits fairly. Its weight is therefore $2^{-v_{\text{old}}} N^{-1} 2^{-[v - v_{\text{old}} - (d - K)]} = 2^{-v - K}$. Unused fair completion is identical. This proves (5.1) at the atom level.

For a fixed full cycle family, let m_{new} count all repeats involving the last cycle: internal repeats plus contexts it shares with earlier cycles. Earlier cycles' internal repeats are not counted in m_{new} . If J counts reused queries in the last cycle's open part (with earlier contexts known at time zero), then

$$J + K = m_{\text{new}}. \quad (5.2)$$

Average only the root of the last cycle. A shared context belongs to its closing part with probability d/r ; an internal repeated context contributes with probability at most $2d/r$. Thus

$$\mathbb{E}_{\text{root}} K \leq 2dm_{\text{new}}/r, \quad \mathbb{E}_{\text{root}} J \geq a_r m_{\text{new}}, \quad a_r = 1 - 2d/r.$$

The function $j \mapsto j2^{j-m_{\text{new}}}$ is increasing and convex for $j \geq 0$. Therefore

$$\mathbb{E}_{\text{root}}[J2^{-K}] \geq a_r m_{\text{new}} 2^{-2dm_{\text{new}}/r}. \quad (5.3)$$

On the sparse class, (3.8) or (3.9) bounds the inverse factor by $H_{k,\beta}$. The forced open part has exactly the ordinary $\mu_{\mathbf{a}}$ -plus-uniform-root law; success only restricts it. To make the integration explicit, let A_{sp} be the sparse class, $\mu = \mu_{\mathbf{a},r}$, $\nu = \nu_{\mathbf{a};r}$, and $H = H_{k,\beta}(d)$. Then

$$\begin{aligned} \frac{a_r}{H} \int_{A_{\text{sp}}} m_{\text{new}} d\mu &\leq \int_{A_{\text{sp}}} J2^{-K} d\mu \\ &= \int_{A_{\text{sp}}} J d\nu \leq \mathbb{E}_{\mu_{\mathbf{a}},X} J. \end{aligned} \quad (5.3a)$$

The sparse class and m_{new} are invariant under re-rooting the last cycle, so the first inequality follows by integrating the root-averaged bound (5.3). The last inequality discards a restriction to successful histories; it does not assert that the feedback law after forced closure is ordinary. Using $1 - 2^{-K} \leq K$ now gives

$$\begin{aligned} 0 \leq Q_d(\mathbf{a}, r) - \nu_{\mathbf{a};r}(\Omega) &\leq \frac{2d}{r} \int_{\text{sparse}} m_{\text{new}} d\mu_{\mathbf{a},r} + D_{\mathbf{a},r} \\ &\leq \frac{2dH_{k,\beta}}{r - 2d} \mathbb{E}J + D_{\mathbf{a},r}. \end{aligned} \quad (5.4)$$

Take $r \geq 4d$ and $n = r - d$. Combining (4.3), (5.4), and $\nu_{\mathbf{a};r}(\Omega) \leq Q_d(\mathbf{a})$, yields the key finite recurrence

$$Q_d(\mathbf{a}, r) \leq Q_d(\mathbf{a}) + \frac{2dH_{k,\beta}}{r - 2d} \left[\frac{1}{N} \sum_{h=1}^{n-1} (n - h) Q_d(\mathbf{a}, h) + \frac{2nP}{N} Q_d(\mathbf{a}) \right] + D_{\mathbf{a},r}. \quad (5.5)$$

$D = 0$ for total length below d^4 ; otherwise it is bounded by (3.6).

No long cycle is charged at a shorter cycle's rate. Prior cycles remain under their ordinary finite measure, rather than being incorrectly treated as an unbiased probability law after earlier forced closures.

6 Close the upper induction

6.1 Base: all lengths below $4d$

Let B mean that every prescribed cyclic word is primitive and no two words represent the same necklace. A_d implies B . On B with no repeated contexts, the weighted integrand is exactly one.

For two distinct positions, Appendix A proves the fully wrapped-window bound

$$\mathbb{P}\{B \text{ and their } (d-1)\text{-contexts agree}\} \leq 2^{-(d-1)}.$$

This includes component words shorter than the window and equal component lengths; the event B removes the periodic degeneracies.

A union bound, together with $m \leq L \log_2 L/(2d)$, implies

$$Q_d(\mathbf{r}) \leq 1 + \frac{L^{2k+2}}{N}, \quad \max_i r_i < 4d, \quad L < 4kd. \quad (6.1)$$

6.2 Crude bound

Induct first on the number of cycles k , then strongly on total length L for fixed k . The target crude bound is $Q_d(\mathbf{r}) \leq 2^k$. The base (6.1) holds for large d . When $r = \max_i r_i \geq 4d$, all same- k moments in (5.5) have total length $P + h < L$. The lower-order moment is bounded by 2^{k-1} . Therefore

$$Q_d(\mathbf{a}, r) \leq 2^{k-1} + C_k d H_{k,\beta}(d) \frac{L}{N} + D_{\mathbf{a},r}.$$

The extra terms tend to zero uniformly for $L \leq N^\beta$, since H is polynomial and $\beta < 1$. Thus they fit within the fixed gap from 2^{k-1} to 2^k . This closes the induction, including moment lists that contain very short components. In particular, $Q_d(h) \leq 2$ for all $h \leq N^\beta$.

This is a strong induction on an integer total length, within an outer induction on the fixed order. For a fixed target order k , take d large enough for all orders at most k . At each non-base step the strictly smaller-order term is bounded by 2^{k-1} , while the same-order terms have strictly smaller total. Thus there is a fixed gap available for the uniformly vanishing error; no unproved sharp estimate is used to initialize the recurrence.

6.3 Sharp constant

Substitute the crude bounds back into (5.5):

$$Q_d(\mathbf{a}, r) \leq Q_d(\mathbf{a}) + C_{k,\beta} d H_{k,\beta}(d) \frac{L}{N} + D_{\mathbf{a},r}.$$

Repeatedly remove a longest component. There are at most k steps. The base (6.1) costs at most a polynomial times $1/N$, absorbed by $P_{k,\beta}(d)L/N$. For $L \geq d^4$ the dense error is at most $C_\beta 2^{-d^3}$, also absorbed. This proves (1.1).

7 A matching lower bound from several middle-first bridges

Only the one-cycle bound $Q_d(h) \leq 2$ is needed in this section. The restriction $L \leq N^\beta$ is used here only to ensure this bound at the required lengths. Thus the numerical lower-bound constant depends only on k , although the threshold for sufficiently large d may also depend on β .

For an ordinary trajectory of $n \geq d$ steps from a uniform root, let $R_{n,d}$ mean that some query in its final d steps repeats an earlier context. The stopped conjugate-switch identity and a union bound over time separations give

$$\mathbb{P}(R_{n,d}) \leq \frac{d+1}{N} \sum_{h \leq n} Q_d(h) \leq \frac{2(d+1)n}{N}. \quad (7.1)$$

The event of a prior return costs $N^{-1} \sum_{h \leq n} Q_d(h)$; on survival a repeated context is a conjugate encounter, and each separation appears at most d times in the terminal query block. Coupling the last d fresh outputs to fair bits also gives

$$d_{\text{TV}}(\mathcal{L}(X_0, X_n), U_N \otimes U_N) \leq \frac{2(d+1)n}{N}. \quad (7.2)$$

Coordinate reversal conjugates the inverse Thorp permutation to the same model with its feedback arguments reversed, so these estimates hold backwards too.

More explicitly, the one-cycle instance of (4.1a) bounds the stopped conjugate endpoint count by $Q_d(h)$. With a uniform root, a prior return has probability $N^{-1} \sum_{h \leq n} Q_d(h)$. On survival, a pair of equal contexts consists of distinct conjugate states. For each separation h , there are at most d pairs with the later query among the final d queries. This proves (7.1).

For (7.2), at each of the final d steps supply a new independent fair comparison bit. If its context is new, choose the freshly revealed feedback value to make the actual output equal that bit. If the context is old, continue the comparison with its independent bit. The comparison endpoint is a uniform d -bit state independent of the initial root; failure of the coupling requires terminal reuse. This constructs the comparison, rather than conditioning a collection of outputs on the event that all are fresh.

If R_d reverses all d coordinates, direct calculation gives

$$R_d T_f^{-1} R_d = T_{f \circ R_{d-1}}.$$

The transformed feedback table is again uniform and independent, so the same stopped-reuse and endpoint estimates hold backwards.

7.1 Reference construction

Assume all $r_i \geq 4d$ and put $t_i = r_i - 2d$. In a common ordinary feedback environment, choose independent uniform roots v_i and run middle paths of t_i steps to z_i . Independently choose uniform w_i . Add the two unique d -step shift paths

$$z_i \longrightarrow w_i \longrightarrow v_i.$$

Let G require all middle states to be distinct, within and between paths, and all $2kd$ added query contexts to be mutually distinct and absent from every middle queried context. On G assign the fresh feedback values to realize the added outputs and complete all remaining values fairly. This produces k disjoint cycles of the required lengths.

7.2 Failure bounds

Internal failure of a middle path costs at most $2t_i/N$.

For different middle paths of lengths t_i, t_j , conditional on any fixed f , an intersection entails

$$v_j = T_f^{a-b} v_i, \quad 0 \leq a \leq t_i, \quad 0 \leq b \leq t_j.$$

There are at most $t_i + t_j + 1$ distinct offsets. Thus

$$\mathbb{P}(\text{middle } i \text{ meets middle } j \mid f) \leq (t_i + t_j + 1)/N. \quad (7.3)$$

There is no $t_i t_j / N$ loss here.

For an added path, separate contacts with its **own** middle from contacts with **other** middles. For self-contacts and contacts with its own middle, consider just that single ordinary middle and its independent seed w_i . Until the first reused query, its forward reference extension agrees with an ordinary extension of that middle, by deferred decisions. Formula (7.1) with $n = t_i + d$ bounds this failure. The other extension is explored backwards from v_i , and the reverse-time version of (7.1) gives the same bound. A backward step from $y = (u, z)$ has predecessor $(z \oplus f(u), u)$ and queries exactly $f(u)$, the same entry as the forward step into y . Hence the reverse-time reuse event controls precisely the relevant forward-context contacts. These are marginal, unconditional estimates; they do not require the feedback law to remain ordinary after conditioning on every other trace.

For contacts with a different middle j , use the reference construction directly. Conditional on f , every fixed context in the added path z_i to w_i is uniform: it uses a suffix of the uniform z_i and a prefix of the independent uniform w_i . The path w_i to v_i has the same property. These contexts are independent of a specified query context in middle j , since (v_i, w_i) and v_j are independent conditional on f . Equality therefore has probability $2/N$. A union bound over at most dt_j pairs for each added path costs $2dt_j/N$. Summing all the self and other-middle bounds costs $O_k(dL/N)$.

For the two added paths belonging to the same cycle, their possible cross contacts depend only on (z_i, w_i, v_i) . By (7.2) this triple is within $2(d+1)t_i/N$ of three independent uniform states. Under the independent law, $z_i w_i v_i$ is a linear independent-bit word; each of the d^2 possible cross-window matches has probability $2/N$, including overlapping windows. This again sums to $O_k(dL/N)$.

For added paths belonging to different cycles, no endpoint approximation is needed. Conditional on f , the pairs (v_i, w_i) are independent across i , z_i is uniform, and each individual added context is uniform. Thus each cross-cycle pair of added contexts agrees with probability $2/N$. There are $O(k^2 d^2)$ such pairs, costing $O(k^2 d^2 / N)$, absorbed into $O_k(dL/N)$.

All estimates are unconditional union bounds; the failure events are not assumed independent. Together,

$$\mathbb{P}_{\text{ref}}(G) \geq 1 - C_k dL/N. \quad (7.4)$$

7.3 Exact good-history weight

Fix the middle traces, whose queried contexts number v_M , and the words w_i . The ordinary root-and-middle probability contains $N^{-k} 2^{-v_M}$. On G the $2kd$ prescribed fresh outputs have probability N^{-2k} ; scaling the ordinary law by N^k leaves weight $2^{-(v_M + 2kd)}$. The reference chooses all w_i with probability N^{-k} , giving the same weight $N^{-2k} 2^{-v_M}$.

The reference construction only retains feedback values queried by the middles. Values at added contexts are assigned after choosing the bridges, and the rest are freshly completed. It does not require an initially sampled unused feedback value to equal a forced value; those initially unused

values have been marginalized. This distinction is part of the weight calculation, not an assumption about their conditional distribution.

Thus the scaled ordinary cycle measure equals the reference measure on G , including the same uniform completion of unused feedback values. Therefore

$$Q_d(\mathbf{r}) \geq \mathbb{P}_{\text{ref}}(G) \geq 1 - C_k dL/N.$$

This proves the lower half of (1.2). It needs neither conditional independence of endpoints given all other traces nor a joint-collision moment theorem.

8 Fixed-order moments and the truncated Gaussian field

Fix $0 < \alpha < 2/3$ and choose β strictly between α and $2/3$. Let $b = \lfloor N^\alpha \rfloor$ and $\ell = \log N$. For every fixed moment order m and all sufficiently large d , all lists of at most m lengths bounded by b have total length $\leq N^\beta$. The moment comparison in this section is initially restricted to $4d \leq r \leq b$; the omitted shorter lengths are restored after the tightness argument. On this restricted range, (1.2) gives

$$\max_{1 \leq j \leq m} \sup_{4d \leq r_1, \dots, r_j \leq b} |Q_d(r_1, \dots, r_j) - 1| \leq \varepsilon_{d,m}, \quad \varepsilon_{d,m} = d^{O_{m,\beta}(1)} N^{\alpha-1}. \quad (8.1)$$

This is smaller than every negative power of d for fixed m .

Let Z_r , $4d \leq r \leq b$, be independent Poisson variables of means $1/r$. Their normalized factorial moments Q are exactly one. Ordinary moments of additive cycle statistics expand over partitions of their selected cycle occurrences into groups that refer to the same cycle. Each block contributes one harmonic summation. For logarithmic weights, a block of multiplicity a contributes at most

$$\sum_{r \leq b} (\log r)^a / r = O_a(\ell^{a+1}).$$

Consequently the error in a fixed raw m -th moment of the logarithmic product is $O_m(\varepsilon_{d,m} \ell^{2m})$. Centering and Gaussian normalization multiply this only by further powers of ℓ . All normalized moment errors vanish. The same argument applies to arbitrary finite collections of count and log-product increments. The corresponding Poisson statistics have Gaussian limits and moment-determinate limiting distributions.

For completeness, the reference Gaussian limit itself needs no permutation-order theorem. At positions $t_r = \log r / \ell$, a linear combination of finitely many count and normalized log-product increments has centered Poisson form

$$\frac{1}{\sqrt{\ell}} \sum_{4d \leq r \leq b} g(t_r)(Z_r - 1/r),$$

where g is a bounded, piecewise continuous function. Its variance converges to $\int_0^\alpha g(t)^2 dt$ by harmonic Riemann sums. Its cumulant of order $j \geq 3$ is

$$\ell^{-j/2} \sum_{4d \leq r \leq b} \frac{g(t_r)^j}{r} = O_j(\ell^{1-j/2}) = o(1).$$

Thus its moments converge to Gaussian moments. The Gaussian law is moment-determinate; the multivariate statement follows by the Cramér–Wold argument. The log-product coordinate corresponds to the weight $\sqrt{3}t$, so the common limiting Brownian motion gives $(W(u), \sqrt{3} \int_0^u t dW(t))$.

Corollary 8.1 (Bounded additive tests). *Fix $0 < \alpha < 2/3$, put $\ell = \log N$, and let $g_1, \dots, g_q$ be a fixed finite family of bounded piecewise-continuous real functions on $[0, \alpha]$. Then*

$$\left(\frac{1}{\sqrt{\ell}} \sum_{4d \leq r \leq N^\alpha} g_j \left(\frac{\log r}{\ell} \right) (C_r - 1/r) \right)_{j=1}^q \implies \mathcal{N}_q(0, \Sigma), \quad \Sigma_{ij} = \int_0^\alpha g_i(t) g_j(t) dt.$$

Proof. Apply the preceding factorial-moment comparison to each linear combination of the g_j . The Poisson reference has variance converging to the stated integral and cumulants of every fixed order $m \geq 3$ of size $O_m(\ell^{1-m/2})$. Its normalized moments converge to Gaussian moments; the comparison transfers them. Cramér–Wold gives the joint conclusion, including singular covariance matrices. $\square$

8.1 Tightness, not just finite-dimensional convergence

First omit lengths below $4d$. Center by the exact Poisson-reference harmonic sums. In the log-product coordinate use weights $\sqrt{3} \log r / \ell$, so both coordinates are sums with bounded weights, divided by $\sqrt{\ell}$.

Extend each truncated centered process constantly beyond α to the next point of the full grid $u = j/\ell$. Interpolate on this full grid and then restrict to $[0, \alpha]$. A grid interval containing $m \geq 1$ full mesh steps has reference Poisson intensity $O(m)$. For either normalized coordinate its fourth centered Poisson moment is at most

$$\frac{C(m^2 + m)}{\ell^2}.$$

Expansion into raw moments of orders at most four and (8.1) gives the same bound for the actual centered coordinate, plus a uniform remainder R_d that is smaller than every power of $1/\ell$. For example, the crude bound $R_d = O(\varepsilon_{d,4}(1 + \ell)^6)$ suffices. Since $m \geq 1$, absorb this remainder into a constant multiple of $(m/\ell)^2$.

Linearly interpolate the grid values. Within a grid cell an increment is a fraction θ of the grid increment. Its fourth moment is at most $C\theta^4/\ell^2 \leq C|u - v|^2$, since $\theta = \ell|u - v| \leq 1$. Across cells split an interval into its first partial cell, full grid part, and final partial cell. The inequality $|x + y + z|^4 \leq 27(|x|^4 + |y|^4 + |z|^4)$ gives the uniform continuity bound $C|u - v|^2$. The constant extension makes the final mesh cell a full cell as well; no uniform estimate for an arbitrarily shortened interpolation cell is needed. The fourth-moment tightness criterion therefore applies to the interpolated processes.

To compare with the original step processes, let D_j count all cycles in one logarithmic grid bin. Its harmonic intensity is bounded by an absolute constant. The upper moment theorem at orders at most four gives $\mathbb{E}D_j^4 \leq C$, uniformly over the bins. There are $O(\ell)$ bins, so for every $\eta > 0$,

$$\mathbb{P}\{\max_j D_j > \eta\sqrt{\ell}\} \leq \frac{C}{\eta^4 \ell} \longrightarrow 0.$$

Within a bin, the normalized log-product oscillation is at most a fixed multiple of $D_j/\sqrt{\ell}$, because $\log r/\ell \leq \alpha$. The deterministic reference centerings have oscillation $O(1/\sqrt{\ell})$ there. Thus both step processes are uniformly close in probability to the tight interpolations.

Below length $4d$, $Q_d(r) \leq 2$ implies expected count $O(\log d)$ and expected log-product $O((\log d)^2)$. Both vanish at their respective normalizations. This also handles the initial interval near $u = 0$. Centering by exact harmonic sums or by the displayed polynomial centerings differs by a uniformly negligible quantity.

Therefore, with

$$K_d(u) = \sum_{r \leq N^u} C_r, \quad B_d(u) = \sum_{r \leq N^u} C_r \log r,$$

we obtain convergence in $D([0, \alpha], \mathbb{R}^2)$, with continuous limiting paths:

$$\left(\frac{K_d(u) - u\ell}{\sqrt{\ell}}, \quad \frac{B_d(u) - \frac{1}{2}u^2\ell^2}{\sqrt{\ell^3/3}} \right) \Rightarrow \left(W(u), \sqrt{3} \int_0^u t dW(t) \right). \quad (8.2)$$

This argument does not require a total-variation approximation of the full initial cycle-count vector. Fixed-order estimates with superpolynomially small errors suffice for the moment and tightness argument. Section 10 separately obtains total-variation approximation on sets of bounded harmonic mass.

9 Product to LCM

For all $r, s \leq b$, the all-positive-length upper theorem gives, for large d ,

$$\mathbb{E}C_r \leq 2/r, \quad \mathbb{E}[C_r(C_s - \mathbf{1}_{r=s})] \leq 2/(rs).$$

Let $D_m = \sum_{m|r, r \leq b} C_r$. The exact arithmetic identity is

$$B_d(\alpha) - \log \text{lcm}\{r \leq b : C_r > 0\} = \sum_{m \leq b} \Lambda(m)(D_m - 1)_+.$$

The preceding harmonic bounds imply

$$\mathbb{E}(D_m - 1)_+ \leq \min \left\{ \frac{2H_{\lfloor b/m \rfloor}}{m}, \frac{H_{\lfloor b/m \rfloor}^2}{m^2} \right\}.$$

Put $t = \log b$. The elementary Chebyshev bound $\psi(x) = \sum_{m \leq x} \Lambda(m) = O(x)$ and partial summation give

$$\sum_{m \leq t} \frac{\Lambda(m)}{m} = O(\log t), \quad \sum_{m > t} \frac{\Lambda(m)}{m^2} = O(1/t). \quad (9.1)$$

For completeness, the product of primes p for which some power satisfies $n < p^a \leq 2n$ divides $\binom{2n}{n}$, and there is at most one such power for each prime. Thus $\psi(2n) - \psi(n) \leq \log \binom{2n}{n} \leq 2n \log 2$; summing over dyadic intervals proves $\psi(x) = O(x)$. In the harmonic bound above, $H_{\lfloor b/m \rfloor} = O(t)$ uniformly. Splitting at $m = t$ therefore yields

$$\begin{aligned} \mathbb{E}(B_d(\alpha) - O_d(\alpha)) &\ll t \sum_{m \leq t} \frac{\Lambda(m)}{m} + t^2 \sum_{m > t} \frac{\Lambda(m)}{m^2} \\ &= O(\log b \log \log b) = o(\ell^{3/2}). \end{aligned} \quad (9.2)$$

The discrepancy is nondecreasing with the cutoff, so the bound is uniform on $0 \leq u \leq \alpha$. Markov's inequality transfers (8.2) from $B_d(u)$ to

$$O_d(u) = \log \text{lcm}\{r \leq N^u : C_r > 0\}.$$

This proves Theorem 1.2, including its scalar consequence.

This is a truncated theorem. The dense-tail estimate used above requires $\kappa > 0$ in (3.5); the argument does not establish the full permutation-order limit.

Corollary 9.1 (Lower bounds for the full permutation). *For every $\varepsilon > 0$,*

$$\mathbb{P}\left\{\log \text{ord}(T_f) \geq (2/9 - \varepsilon)(\log N)^2\right\} \longrightarrow 1, \quad \mathbb{P}\left\{\sum_{r=1}^N C_r \geq (2/3 - \varepsilon) \log N\right\} \longrightarrow 1.$$

Proof. For each fixed $\alpha < 2/3$, Theorem 1.2 gives the laws of large numbers $O_d(\alpha)/\ell^2 \rightarrow \alpha^2/2$ and $K_d(\alpha)/\ell \rightarrow \alpha$ in probability. The full order and count dominate their truncated counterparts. Choose a fixed α sufficiently close to $2/3$. No uniform estimate as $\alpha \uparrow 2/3$ is used. $\square$

10 Poisson limits and further cycle laws

The moment theorem gives a growing-dimensional total-variation result when the total harmonic mass of the observed lengths is bounded. This is different from approximating the entire initial vector through a power of N .

Theorem 10.1 (Bounded harmonic mass). *Fix $0 < \alpha < 2/3$ and $\Lambda < \infty$. Let $I_d \subseteq \{r : 4d \leq r \leq N^\alpha\}$ satisfy $\lambda_d = \sum_{r \in I_d} r^{-1} \leq \Lambda$. Then*

$$d_{\text{TV}}\left(\mathcal{L}((C_r)_{r \in I_d}), \bigotimes_{r \in I_d} \text{Poisson}(1/r)\right) \longrightarrow 0. \quad (10.1)$$

The cardinality of I_d may tend to infinity.

10.1 Finite inclusion–exclusion

We give a finite comparison that makes the order of limits explicit. For a vector $X = (X_i)_{i \in I}$ of nonnegative integers on a finite set, write its ordered factorial moment measures as

$$\mu_j(i_1, \dots, i_j) = \mathbb{E} \prod_{i \in I} (X_i)_{a_i}, \quad a_i = \#\{v : i_v = i\}.$$

Let Y_i be independent Poisson variables with means λ_i , put $\lambda = \sum_i \lambda_i$, and let $\pi_j(i_1, \dots, i_j) = \prod_v \lambda_{i_v}$. The norm $\|\cdot\|_1$ below is total absolute mass on I^j .

Lemma 10.2 (Finite factorial-moment comparison). *If $\mu_p(I^p) < \infty$ for an integer $p \geq 1$, then*

$$\begin{aligned} d_{\text{TV}}(\mathcal{L}(X), \mathcal{L}(Y)) &\leq \sum_{j=1}^{p-1} \frac{2^j}{j!} \|\mu_j - \pi_j\|_1 \\ &\quad + \frac{2^p}{p!} (\mu_p(I^p) + \lambda^p). \end{aligned} \quad (10.2)$$

Proof. Put $M = \sum_i X_i$ and take $0 \leq F \leq 1$ on configurations. For $q < p$, use the symmetric test $F_q(i_1, \dots, i_q) = F(\sum_{v=1}^q \delta_{i_v})$ on ordered distinct *occurrences*; occurrences at the same location are still distinct. After selecting q occurrences, approximate the event that no others remain by

$$\sum_{j=0}^{p-q-1} \frac{(-1)^j}{j!} (M - q)_j.$$

For each integer $n \geq 0$, the error in this approximation to $\mathbf{1}_{\{n=0\}}$ is at most $(n)_{p-q}/(p-q)!$. Indeed, when $n > 0$ the partial binomial sum is $(-1)^{p-q-1} \binom{n-1}{p-q-1}$; it vanishes when the cutoff exceeds n , and otherwise its absolute value is at most $\binom{n}{p-q}$.

Divide by $q!$ to compensate for ordering the selected occurrences. The error for this q is at most $\mu_p(I^p)/(q!(p-q)!)$. The omitted event $M \geq p$ costs at most $\mu_p(I^p)/p!$. Summing over $q = 0, \dots, p-1$ bounds the remainder by $2^p \mu_p(I^p)/p!$. For the Poisson vector the analogous remainder is $2^p \lambda^p/p!$. At total factorial order $j < p$, the sum of absolute coefficients is

$$\sum_{q=0}^j \frac{1}{q!(j-q)!} = \frac{2^j}{j!}.$$

The order-zero terms agree. Subtraction of the two finite expansions and supremization over indicator tests F prove the lemma. $\square$

Proof of Theorem 10.1. Choose $\alpha < \beta < 2/3$. For every fixed j , all ordered lists of j lengths from I_d eventually have total at most N^β . Theorem 1.1 gives a uniform error $\varepsilon_{d,j} \rightarrow 0$ and hence

$$\|\mu_j - \pi_j\|_1 \leq \varepsilon_{d,j} \lambda_d^j, \quad \mu_j(I_d^j) \leq (1 + \varepsilon_{d,j}) \lambda_d^j.$$

Substitution in (10.2) yields, for fixed p ,

$$d_{\text{TV}} \leq \sum_{j=1}^{p-1} \frac{(2\Lambda)^j}{j!} \varepsilon_{d,j} + \frac{(2\Lambda)^p}{p!} (2 + \varepsilon_{d,p}). \quad (10.3)$$

First let $d \rightarrow \infty$ with p fixed, and then let $p \rightarrow \infty$. No estimate uniform in growing moment order is required. $\square$

10.2 The scale-invariant cycle process

The scale-invariant Poisson process has intensity dx/x on $(0, \infty)$; see Arratia [1] for its role in permutation cycles and other logarithmic structures. The next result identifies this process as a moving mesoscopic limit for the present feedback model.

Corollary 10.3 (Scale-invariant Poisson limit). *Let $a_d/d \rightarrow \infty$, and assume $a_d \leq N^\alpha$ for some fixed $0 < \alpha < 2/3$. Then the random counting measures*

$$\Xi_d = \sum_{r=1}^N C_r \delta_{r/a_d} \implies \Xi \sim \text{PPP}(dx/x) \quad (10.4)$$

converge in distribution for the vague topology on locally finite counting measures on $(0, \infty)$.

Proof. On a compact interval $[c, C] \subset (0, \infty)$, the relevant lengths exceed $4d$ eventually and are at most CN^α . Use a slightly larger fixed exponent below $2/3$ in Theorem 10.1. Their harmonic mass is bounded, and the discrete measures $\sum_r r^{-1} \delta_{r/a_d}$ converge vaguely to dx/x . The Laplace functionals of the corresponding Poisson processes therefore converge. Total variation transfers this convergence for compactly supported tests. Bounded expected counts on each compact interval give local tightness, proving the assertion. $\square$

In particular, for $b = \lfloor N^\alpha \rfloor$ with $0 < \alpha < 2/3$,

$$\sum_{b < r \leq 2b} C_r \implies \text{Poisson}(\log 2).$$

A fixed finite collection of disjoint multiplicative windows has independent limiting counts, with means equal to the logarithms of their endpoint ratios. The windows may be at widely separated scales. These conclusions do not apply to the macroscopic window $(N/2, N]$, which contains at most one cycle.

10.3 Dickman mass and the largest cycle below a cutoff

The relation between scale-invariant Poisson processes and Dickman laws is classical [1, 4]. We give the short transfer proof to specify the normalization and the Thorp application.

Corollary 10.4 (Truncated cycle mass). *Fix $0 < \alpha < 2/3$, put $b = \lfloor N^\alpha \rfloor$, and define*

$$T_d(b) = \frac{1}{b} \sum_{r \leq b} r C_r.$$

Then $T_d(b) \implies D$, where the standard Dickman variable D is specified by

$$\mathbb{E} e^{-tD} = \exp \left\{ \int_0^1 \frac{e^{-tx} - 1}{x} dx \right\}, \quad t \geq 0. \quad (10.5)$$

Equivalently, $D = \int_{(0,1]} x \Xi(dx)$ for the process in Corollary 10.3. The limiting mean and variance are $\mathbb{E}D = 1$ and $\text{Var}(D) = 1/2$.

Proof. For fixed $\varepsilon > 0$, integration on $(\varepsilon, 1]$ is continuous at the limiting point process almost surely: it has no atoms at either boundary. Thus the contribution from $\varepsilon b < r \leq b$ converges by Corollary 10.3. The all-positive-length upper bound gives

$$\mathbb{E} \left[\frac{1}{b} \sum_{r \leq \varepsilon b} r C_r \right] \leq 2\varepsilon$$

for sufficiently large d . The limiting omitted Poisson mass has expectation ε . Markov's inequality and $\varepsilon \downarrow 0$ complete the convergence argument. The Poisson Laplace functional gives (10.5); its first two cumulants are $\int_0^1 x dx/x = 1$ and $\int_0^1 x^2 dx/x = 1/2$. $\square$

Corollary 10.5 (Largest cycle below a cutoff). *With b as above, let $R_d(b) = \max(\{r \leq b : C_r > 0\} \cup \{0\})$. Then*

$$\frac{R_d(b)}{b} \implies \text{Uniform}(0, 1). \quad (10.6)$$

Proof. For $0 < x < 1$, the event $R_d(b)/b \leq x$ is the absence of cycles in $(xb, b]$. Its limiting probability, by Theorem 10.1, is $\exp(-\int_x^1 dt/t) = x$. $\square$

The harmonic mass of the entire range $[4d, N^\alpha]$ diverges. Consequently Theorem 10.1 does not yield full-vector total-variation approximation on that range. The next section obtains an explicit estimate for an initial range below the birthday scale by a separate argument.

11 Quantitative approximation below the birthday scale

This section gives an explicit error for the complete initial cycle-count vector. Its proof uses context-simple necklace indicators, not a growing-order version of Theorem 1.1.

Theorem 11.1 (Quantitative initial-range approximation). *There is an absolute constant C such that, for sufficiently large d and $2d \leq b \leq \sqrt{N}$,*

$$d_{\text{TV}} \left(\mathcal{L}((C_r)_{2d \leq r \leq b}), \bigotimes_{r=2d}^b \text{Poisson}(1/r) \right) \leq C d^4 (\log_2 d)^4 \frac{b^2}{N}. \quad (11.1)$$

In particular, the distance tends to zero when $b = o(\sqrt{N}/[d^2(\log_2 d)^2])$.

11.1 Context-simple cycles and sequential forced closure

A cycle is *context-simple* if its queried length- $(d-1)$ contexts are all distinct. Let C_r^{rep} count the remaining r -cycles, and let $X_{r,s}$ count ordered pairs of distinct context-simple cycles of lengths r, s sharing at least one context. Equal lengths are allowed.

We first control one repeated-context cycle or a pair of context-simple cycles. Throughout this section, use the following forced probability law: choose independent uniform roots and construct each prescribed cycle by $r-d$ ordinary steps and d forced closing steps. For a pair, construct the shorter cycle completely before constructing the longer one. Reject an incompatible closure, a premature return, or an intersection of the cycles. Let V count closing queries already fixed by *any* earlier query, including earlier closing queries. If the completed family uses v contexts, the forced atom weight is 2^{-v-V} . Thus, on a root-invariant class E_m with m repeated contexts and total length L , root averaging gives

$$\mathbb{P}^\circ(E_m) = \int_{E_m} \mathbb{E}_{\text{roots}}(2^{-V}) d\mu. \quad (11.2)$$

Here μ is the corresponding scaled ordinary cycle measure. For one cycle, $\mathbb{E}_{\text{roots}} V \leq 2dm/r$. For a context-simple pair $r \leq s$, only the second cycle can have known closing queries, and $\mathbb{E}_{\text{roots}} V = dm/s \leq 2dm/(r+s)$. Therefore, writing $Q^{(m)}$ for the contribution of E_m ,

$$Q^{(m)} \leq 2^{2dm/L} \mathbb{P}^\circ(E_m). \quad (11.3)$$

Until the first repeated feedback query, this construction agrees with independent fair cyclic seed words: the starting d bits and the $r-d$ open-step output bits are independent, and closing appends the starting bits. For $r \geq 2d$, each pair of distinct $(d-1)$ -contexts on a circle matches with probability $2^{-(d-1)} = 2/N$, by Appendix A. For two different circles the same probability follows from independence. On a successful intersecting context-simple pair, the first repetition must be between the two circles. Hence

$$\begin{aligned} \sum_{m \geq 1} \mathbb{P}^\circ(E_m) &\leq r(r-1)/N \quad \text{for one cycle,} \\ \sum_{m \geq 1} \mathbb{P}^\circ(E_m) &\leq 2rs/N \quad \text{for a context-simple pair.} \end{aligned} \quad (11.4)$$

No unexposed feedback value is resampled after its first query.

11.2 A uniform sparse penalty

For $d^4 \leq L \leq 2\sqrt{N}$, the one- or two-word type estimate is

$$\log_2 Q^{(m)} \leq \frac{Lt}{h} + 2M \log_2[e(1+L/M)] + 2\log_2 L - \kappa m, \quad M = 2^h, \quad (11.5)$$

where $t = \log_2 L - h$ and $\kappa = (2d-3h)/h$. Set $q = \log_2 d$ and

$$h = \lfloor \log_2 L - \log_2 d - \log_2 q \rfloor.$$

For sufficiently large d , $2 \leq h \leq d/2$, $\kappa \geq 1$, $L/(2dq) < M \leq L/(dq)$, and $2M \log_2[e(1+L/M)] \leq 4L/d$. Put

$$F = Lt/h + 4L/d + 2\log_2 L, \quad m_0 = \left\lceil \frac{F + L/d}{\kappa} \right\rceil.$$

Then

$$\sum_{m \geq m_0} Q^{(m)} \leq 2^{1-L/d} \leq 2^{1-d^3}. \quad (11.6)$$

For $m < m_0$, the penalty in (11.3) is at most $\Pi = 2^{2dm_0/L}$. Direct expansion gives

$$\log_2 \Pi \leq \frac{2d}{2d-3h} \left[t + \frac{5h}{d} + \frac{2h \log_2 L}{L} \right] + \frac{2d}{L} \leq 4t + 15.$$

Here the leading factor is at most 4, $5h/d \leq 5/2$, $2h \log_2 L/L \leq 1$, and $2d/L \leq 1$. Since $t < q + \log_2 q + 1$, one may take

$$\Pi \leq 2^{20} d^4 q^4. \quad (11.7)$$

Combining (11.3)–(11.7) yields

$$\begin{aligned} \mathbb{E}C_r^{\text{rep}} &\leq \Pi \frac{r-1}{N} + \frac{2^{1-r/d}}{r} \quad (r \geq d^4), \\ \mathbb{E}X_{r,s} &\leq \frac{2\Pi}{N} + \frac{2^{1-(r+s)/d}}{rs} \quad (r+s \geq d^4). \end{aligned} \quad (11.8)$$

For the remaining short ranges, the following elementary bounds suffice:

$$\mathbb{E}C_r^{\text{rep}} \leq \frac{r(r-1)}{N}, \quad \mathbb{E}X_{r,s} \leq \frac{4\sqrt{rs}}{N}. \quad (11.9)$$

For one cycle, $H_d = \log_2 r$ and $H_d - H_{d-1} = 2m/r$ imply $2dm/r \leq \log_2 r$. Jensen in (11.2) therefore loses at most a factor r ; (11.4) proves the first bound. For a context-simple pair, choose one of the two circles with probability $1/2$, then a position uniformly on that circle. This stationary mixture has $H_d = 1 + \frac{1}{2} \log_2(rs)$. Each shared context merges atoms of masses $1/(2r)$ and $1/(2s)$ and contributes at least $1/s$ to the final entropy increment. Thus $dm/s \leq 1 + \frac{1}{2} \log_2(rs)$, and the inverse root-averaged likelihood is at most $2\sqrt{rs}$. The second bound follows from (11.4) after dividing by rs .

Summing (11.8) and (11.9) gives

$$\sum_{r=2d}^b \mathbb{E}C_r^{\text{rep}} + \sum_{r,s=2d}^b \mathbb{E}X_{r,s} \leq C d^4 q^4 \frac{b^2}{N}. \quad (11.10)$$

Indeed, on $r < d^4$ use $r^2 \leq d^4 r$, and on $r+s < d^4$ use $\sqrt{rs} \leq d^4/2$. The dense remainders are at most a polynomial in b times 2^{-d^3} and are negligible uniformly for $b \leq \sqrt{N}$.

11.3 The point-process comparison

Let $\mathcal{H}_r$ be the set of possible context-simple primitive necklaces of length r . The event that $\gamma \in \mathcal{H}_r$ occurs as a cycle has indicator I_γ and probability $p_\gamma = 2^{-r}$. Its dependency neighborhood consists of all potential necklaces sharing one of its contexts, including itself. The indicator is independent of the joint collection outside that neighborhood, since their determining feedback bits are disjoint.

The point-process theorem of Arratia, Goldstein, and Gordon [3, Theorem 2], in our total-variation convention, bounds the distance from the independent Poisson indicators by $2(b_1 + b_2)$, where

$$b_1 = \sum_{\gamma} \sum_{\gamma' \sim \gamma} p_\gamma p_{\gamma'}, \quad b_2 = \sum_{\gamma} \sum_{\substack{\gamma' \sim \gamma \\ \gamma' \neq \gamma}} \mathbb{E}(I_\gamma I_{\gamma'}).$$

The external-dependence term is zero. For each pair of lengths r, s , rooting both necklaces gives

$$\sum_{\substack{\gamma \in \mathcal{H}_r, \gamma' \in \mathcal{H}_s \\ \gamma \sim \gamma'}} p_\gamma p_{\gamma'} \leq \frac{1}{rs} \mathbb{P}\{\text{two independent seed words share a context}\} \leq \frac{2}{N}.$$

This includes the diagonal $\gamma = \gamma'$. Consequently $b_1 \leq 2b^2/N$. Simultaneously occurring distinct necklaces are distinct actual cycles, so (11.10) gives $b_2 \leq \sum_{r,s} \mathbb{E}X_{r,s} = O(d^4 q^4 b^2/N)$. Aggregation by length cannot increase total variation.

Let $\lambda_r = |\mathcal{H}_r|2^{-r}$ be the mean context-simple cycle count. A length- r cyclic seed has all contexts distinct if and only if it is a rooted context-simple necklace. Therefore

$$r\lambda_r = 1 - \delta_d(r), \quad 0 \leq \frac{1}{r} - \lambda_r \leq \frac{r-1}{N},$$

where $\delta_d(r) \leq r(r-1)/N$ is its context-collision probability. Changing the independent Poisson means to $1/r$ costs $O(b^2/N)$. Restoring non-context-simple cycles costs at most $\sum_r \mathbb{E}C_r^{\text{rep}}$, bounded in (11.10). This proves Theorem 11.1.

Theorems 10.1 and 11.1 are complementary. The former applies beyond the birthday scale but only to sets of bounded harmonic mass, with no asserted rate; the latter controls the complete initial vector and gives an explicit rate. Neither asserts a Poisson approximation for that complete vector above the birthday scale.

12 Scope and further directions

The parameter restrictions are fixed before $d \rightarrow \infty$. Theorem 1.1 is uniform in length ratios but not in a growing number of cycles. The functional theorem concerns cycles of length at most N^α , for fixed $\alpha < 2/3$, rather than the full order of T_f . The Poisson theorem on bounded-harmonic-mass sets does allow a growing number of coordinates, but it does not cover the whole initial vector when its harmonic mass diverges.

The full one-shuffle Erdős–Turán statement suggested by these results is

$$\frac{\log \text{ord}(T_f) - \frac{1}{2}(\log N)^2}{\sqrt{(\log N)^3/3}} \implies \mathcal{N}(0, 1).$$

It is not proved here. The present dense-tail estimate requires $\kappa = (2d - 3h)/h > 0$. Extending the moment comparison to the remaining length scales therefore requires an additional estimate, not just taking β to the endpoint in a fixed-parameter theorem.

The lower and upper moment arguments are analytic and combinatorial. Finite integer-arithmetic checks of feedback switches, root weights, entropy inequalities, and bridge identities accompany the source. They are reproducibility aids, not inputs to the asymptotic proof. The separate repository documentation records the checks, review history, and their limitations.

A Equality graphs and exact short cycles

A.1 Wrapped-context comparisons

This appendix proves the small-length collision estimate used in (6.1). Let B require primitive component words and different necklaces for any two components of the same length.

For two positions on the same circle of length n , with nonzero displacement a , the h consecutive comparisons are edges $t \leftrightarrow t + a$ modulo n . Write $g = \gcd(n, a)$. For $h < n$, the omitted $n - h$ starting positions are consecutive and meet $\min(g, n - h)$ of the g cycles of the full shift graph. The retained equality system therefore has rank

$$h - \max(0, g - n + h) = \min(h, n - g)$$

over $\mathbb{F}_2$. The same formula holds for $h \geq n$, when the rank has already saturated. If $h \geq n - g$, equality forces period dividing $g < n$, which is impossible on B . Otherwise the rank is h , so equality has probability 2^{-h} . This includes a two-cycle in the shift graph: its two oriented edges impose one linear relation.

For positions on different circles of lengths r, s , offset their roots so that the comparisons are $(t \bmod r) \leftrightarrow (t \bmod s)$, $0 \leq t < h$, in a bipartite graph. Let $g = \gcd(r, s)$ and assume $r \leq s$. Its first s edges attach each vertex of the second circle once. Contract those edges. The next $r - g$ edges become $j \leftrightarrow j + s \pmod{r}$, $0 \leq j < r - g$. The full shift-by- s graph has g cycles, and the omitted g consecutive edges remove exactly one from each. Thus the first $r + s - g$ comparison edges form a forest and have that rank. At this threshold both words are forced to have period dividing g . This is excluded by B : either a component is nonprimitive, or the two have equal length and represent the same necklace. For smaller h the comparisons have rank h and probability 2^{-h} .

Taking $h = d - 1$ proves, for every pair of distinct positions,

$$\mathbb{P}\{B \text{ and their contexts agree}\} \leq 2^{-(d-1)}.$$

On B without such a collision, the weighted-word integrand equals one. On valid words its maximum is $2^m \leq L^{L/(2d)} \leq L^{2k}$ when every length is below $4d$. The union bound over $\binom{L}{2}$ pairs gives (6.1).

A.2 Exact short-cycle laws

Let

$$\eta_r = \frac{1}{r} \sum_{a|r} \mu(a) 2^{r/a}$$

be the number of primitive binary necklaces of length r , where μ is the Möbius function. The independent-binomial phenomenon was proved by Stark [7]. The wrapped-context comparison above yields the following explicit range directly.

Proposition A.1 (Independent short-cycle counts). *For $1 \leq r \leq d - 1$,*

$$C_r \sim \text{Binomial}(\eta_r, 2^{-r}).$$

For $d \geq 2$, the variables $C_1, \dots, C_{\lfloor (d+1)/2 \rfloor}$ are mutually independent.

Proof. A primitive necklace of length $r \leq d - 1$ has r distinct contexts and occurs with probability 2^{-r} . Distinct necklaces of the same length have disjoint context sets: agreement on $d - 1 \geq r$ consecutive positions would identify their periodic extensions. This gives the binomial marginal.

Put $b = \lfloor (d + 1)/2 \rfloor$. For different lengths $r, s \leq b$,

$$r + s - \gcd(r, s) \leq 2b - 2 \leq d - 1.$$

The two-circle comparison above shows that a common context would identify the periodic extensions, contradicting their distinct minimal periods. Thus every primitive-necklace indicator involved in these lengths depends on a disjoint set of feedback bits. This proves joint independence. $\square$

Stark’s stated joint range is $r \leq (d - 1)/3$; the proposition records the larger elementary range furnished by the present equality-graph argument. In particular $C_1 \sim \text{Binomial}(2, 1/2)$, so a Poisson approximation to all fixed small lengths cannot hold. This explains the lower cutoff in the harmonic moment asymptotics.

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